

ISHINOMAKI, Japan

WMO: 475920

Lat: 38.4275N

Lon: 141.2994E

Elev: 43

StdP: 100.81

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.8	(c) -3.8	(d) -9.9	(e) 1.6	(f) -1.4	(g) -8.9	(h) 1.8	(i) -0.9	(j) 12.8	(k) 1.3	(l) 11.8	(m) 2.0	(n) 4.0	(o) 330	(p) 0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.3	(c) 29.5	(d) 23.7	(e) 28.1	(f) 23.4	(g) 27.0	(h) 23.1	(i) 24.9	(j) 27.7	(k) 24.4	(l) 27.0	(m) 23.8	(n) 26.3	(o) 4.5	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.1	19.1	26.4	23.5	18.4	26.1	22.9	17.8	25.7	76.3	27.9	74.0	27.1	71.8	26.4	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.7	(b) 10.2	(c) 9.0	(e) -6.8	(f) 32.9	(g) 1.4	(h) 1.6	(i) -7.8	(j) 34.0	(k) -8.6	(l) 35.0	(m) -9.4	(n) 35.8	(o) -10.4	(p) 37.0		
(4) 11.7	(5) 10.2	(6) 9.0	(7) -6.8	(8) 32.9	(9) 1.4	(10) 1.6	(11) -7.8	(12) 34.0	(13) -8.6	(14) 35.0	(15) -9.4	(16) 35.8	(17) -10.4	(18) 37.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	1.1	1.6	4.6	9.6	14.8	18.4	22.4	24.1	20.8	15.0	8.8	3.6	(6)
(7)		DBStd	8.41	2.37	2.57	2.81	3.15	2.78	2.47	2.77	2.44	2.90	2.97	3.21	2.82	(7)
(8)		HDD10.0	986	276	235	168	44	2	0	0	0	2	60	199		(8)
(9)		HDD18.3	2693	534	468	424	261	113	28	2	0	9	109	286	457	(9)
(10)		CDD10.0	1765	0	0	2	33	151	253	385	437	323	157	24	1	(10)
(11)		CDD18.3	431	0	0	0	0	4	32	129	178	82	6	0	0	(11)
(12)		CDH23.3	1998	0	0	0	1	12	58	568	1043	308	8	0	0	(12)
(13)		CDH26.7	329	0	0	0	0	2	6	99	191	32	0	0	0	(13)
(14)	Wind	WSAvg	4.2	4.6	4.7	4.9	4.7	4.3	3.7	3.4	3.5	3.7	4.0	4.2	4.5	(14)
(15)	Precipitation	PrecAvg	1119	39	35	69	90	100	121	151	118	153	127	72	45	(15)
(16)		PrecMax	1463	132	83	159	169	203	266	321	468	446	385	157	181	(16)
(17)		PrecMin	825	5	6	6	8	32	56	50	15	12	26	14	11	(17)
(18)		PrecStd	169	33	21	39	45	42	46	76	103	94	86	44	35	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	11.5	15.6	20.8	24.9	27.2	31.5	32.4	29.4	24.3	18.2	14.2	(19)
(20)			MCWB	6.4	7.5	9.4	12.5	17.2	19.8	24.3	24.6	22.9	18.6	13.8	11.3	(20)
(21)		2%	DB	7.8	9.1	12.7	18.0	22.1	24.9	29.1	30.2	27.4	22.3	16.9	11.6	(21)
(22)			MCWB	4.8	5.9	8.5	12.1	15.2	19.5	23.6	24.1	22.7	18.3	13.3	8.6	(22)
(23)		5%	DB	6.4	7.6	11.1	16.1	20.3	23.3	27.5	28.6	26.2	21.0	15.8	9.9	(23)
(24)			MCWB	3.5	4.4	7.4	11.2	14.9	18.9	23.2	23.9	22.5	17.3	12.6	6.9	(24)
(25)		10%	DB	5.1	6.2	9.7	14.6	18.9	22.0	26.2	27.5	25.1	19.9	14.6	8.4	(25)
(26)			MCWB	2.4	3.2	6.5	10.6	14.6	18.5	22.8	23.7	22.0	16.4	11.5	5.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	8.6	11.1	14.9	18.8	21.9	25.3	26.0	24.9	21.1	15.6	11.9	(27)
(28)			MCDWB	9.3	10.6	13.4	17.7	21.9	24.7	28.9	29.0	27.1	22.6	16.9	13.6	(28)
(29)		2%	WB	5.2	6.4	9.6	13.5	17.3	20.7	24.5	25.2	24.1	19.4	14.4	9.0	(29)
(30)			MCDWB	7.3	8.7	11.8	16.2	19.6	23.3	27.6	28.0	26.2	21.3	16.1	11.1	(30)
(31)		5%	WB	3.8	4.9	8.2	12.4	16.4	19.9	23.9	24.7	23.4	18.1	13.2	7.3	(31)
(32)			MCDWB	6.0	7.3	10.4	14.9	18.8	22.0	26.6	27.4	25.6	20.2	15.4	9.7	(32)
(33)		10%	WB	2.6	3.4	6.9	11.4	15.5	19.1	23.2	24.2	22.5	16.9	11.8	5.7	(33)
(34)			MCDWB	4.8	5.8	9.5	13.8	17.9	21.3	25.6	26.8	24.7	19.3	14.1	8.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.4	6.9	7.6	7.9	6.8	5.6	5.1	5.3	6.1	7.5	7.9	6.7	(35)
(36)			MCDBR	8.1	8.4	9.5	9.9	9.1	7.5	7.2	7.1	6.7	7.7	9.2	8.6	(36)
(37)			MCWBR	6.0	6.2	6.8	5.7	4.7	3.7	3.2	3.1	3.5	4.9	6.7	6.7	(37)
(38)		5% WB	MCDBR	8.0	8.2	9.0	8.3	7.2	6.0	6.1	5.8	5.4	6.5	8.6	8.7	(38)
(39)			MCWBR	6.5	6.7	7.2	5.7	4.5	3.4	3.0	2.8	3.2	5.1	7.2	7.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.365	0.409	0.483	0.531	0.552	0.545	0.554	0.540	0.475	0.435	0.398	0.373	(40)	
(41)		taud	2.192	2.067	1.911	1.852	1.865	1.961	2.000	2.030	2.178	2.187	2.200	2.197	(41)	
(42)		Ebn at Noon	783	792	768	759	752	757	745	744	772	760	739	740	(42)	
(43)		Edh at Noon	106	138	178	200	201	183	175	166	135	121	105	99	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.11	2.86	3.78	4.69	5.04	4.87	4.44	4.39	3.64	2.99	2.23	1.80	(44)	
(45)		RadStd	0.12	0.20	0.22	0.45	0.49	0.30	0.67	0.57	0.34	0.30	0.14	0.13	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+0.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.38	N/A	N/A	N/A	N/A

Nomenclature: See separate page